



Figure 1: A browser-based interface for NAS

Table 1

Network Attached Storage (NAS)	Storage Area Network (SAN)
NAS can be mapped as network drives to share on that server	SAN can be used as a 'disk in disk' and volume management utilities.
Provides storage and a file system	Block-based storage
Protocols: NFS, CIFS and SMB	Protocols: SCSI, Fibre Channel, iSCSI, ATA over Ethernet and HyperSCSI
For small and medium businesses	For large organisations
Storage capacity: up to a few TB	Storage capacity: many TB
Multiple NAS devices	Single SAN and multiple high-performance disk arrays
Specialised knowledge and training is not required to configure and maintain NAS	Specialised knowledge and training is required to configure and maintain SANs
SAN and NAS are not mutually exclusive. A hybrid of SAN and NAS can offer file- and block-level protocols from the same system to support NAS and SAN.	

Open source implementations

Open source NAS-oriented distributions of Linux and FreeBSD are available, such as FreeNAS, NAS4Free, CryptoNAS, NASLite, Gluster, OpenFiler, and OpenMediaVault, which can be installed on commodity hardware, connected and configured using a Web browser. Open source NAS implementation can run from virtual machines as well as USB disks; we will look at both these cases in the coming section on FreeNAS installation.



Figure 2: A console-based interface for NAS



Figure 3: USB destination media



Figure 4: FreeNAS installation warning

FreeNAS: An overview

FreeNAS is an advanced FreeBSD-based operating system for network-attached storage servers, with an easy-to-use Web UI. Distributed as an ISO image, it is released under a BSD licence, which imposes nominal limitations on the redistribution of covered software. The FreeNAS Project was initially founded by Olivier Cochard-Labbe in 2005, done in PHP and based on an embedded firewall, which is based on FreeBSD. FreeNAS 8.3.0 is based on FreeBSD 8.3, and so supports the hardware in the FreeBSD 8.3 Hardware Compatibility List. FreeNAS is available for both 32-bit (addresses up to 4 GB of RAM) and 64-bit architectures (for speed and performance).

Features

FreeNAS supports protocols such as NFS, SSH, FTP, TFTP,